

Problem 1) $L = T - U$; $\frac{\partial L}{\partial x_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} = 0$ $i = 1, 2, 3$

$\Rightarrow \frac{\partial (T-U)}{\partial x_i} - \frac{d}{dt} \frac{\partial (T-U)}{\partial \dot{x}_i} = 0$; Recall that for conservative systems,

$U = U(x_i)$; $T = T(\dot{x}_i) \Rightarrow \frac{\partial U}{\partial x_i} = \frac{\partial T}{\partial \dot{x}_i} = 0$

$\Rightarrow -\frac{\partial U}{\partial x_i} = \frac{d}{dt} \frac{\partial T}{\partial \dot{x}_i}$ $\left. \begin{array}{l} \text{For conservative systems } F_i = -\frac{\partial U}{\partial x_i} \end{array} \right\} \Rightarrow \frac{d}{dt} \frac{\partial T}{\partial \dot{x}_i} = \frac{d}{dt} \frac{\partial}{\partial \dot{x}_i} \left(\sum_j \frac{1}{2} m \dot{x}_j^2 \right)$

$= \frac{d}{dt} (m \dot{x}_i) = \dot{p}_i$

$= F_i \Rightarrow \boxed{F_i = \dot{p}_i}$

Problem 2) Let's compute the terms in generalized Lagrange's eqn.

$\frac{\partial L}{\partial \dot{q}_j} = \sum_{i=1}^N \frac{\partial L}{\partial \dot{x}_i} \frac{\partial \dot{x}_i}{\partial \dot{q}_j}$ because: $q_j \equiv q_j(x_1, x_2, \dots, x_N; t)$ (I)

We can rewrite $\dot{x}_i$ using $x_i \equiv x_i(q_1, \dots, q_N; t)$ as,

$\dot{x}_i = \sum_{j=1}^N \frac{\partial x_i}{\partial q_j} \dot{q}_j + \frac{\partial x_i}{\partial t} \Rightarrow \frac{\partial \dot{x}_i}{\partial \dot{q}_j} = \frac{\partial x_i}{\partial q_j}$ (II)

(I), (II) $\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = \sum_{i=1}^N \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) \frac{\partial \dot{x}_i}{\partial \dot{q}_j}$

$+ \sum_{i=1}^N \frac{\partial L}{\partial x_i} \frac{d}{dt} \frac{\partial x_i}{\partial q_j}$ (III)

However, $\frac{d}{dt} \frac{\partial \dot{x}_i}{\partial \dot{q}_j} = \sum_{k=1}^N \frac{\partial}{\partial q_k} \left(\frac{\partial \dot{x}_i}{\partial \dot{q}_j} \right) \dot{q}_k + \frac{\partial}{\partial t} \frac{\partial \dot{x}_i}{\partial \dot{q}_j}$

$= \frac{\partial}{\partial q_j} \left(\sum_{k=1}^N \frac{\partial x_i}{\partial q_k} \dot{q}_k + \frac{\partial x_i}{\partial t} \right)$

$= \frac{\partial \dot{x}_i}{\partial q_j}$ (IV)

We know, $\frac{\partial L}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) = 0$ (V)

(III), (IV), (V) $\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = \sum_{i=1}^N \frac{\partial L}{\partial x_i} \frac{\partial \dot{x}_i}{\partial \dot{q}_j} + \sum_{i=1}^N \frac{\partial L}{\partial \dot{x}_i} \frac{\partial \dot{x}_i}{\partial \dot{q}_j}$

$= \frac{\partial L}{\partial q_j}$

$\Rightarrow \boxed{\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = 0}$

Problem 3) $L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgy$; $\frac{\partial L}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) = 0$

There is no x, y dependence; $\frac{\partial L}{\partial \dot{x}} = m\dot{x} \equiv \text{const}$; $\frac{\partial L}{\partial \dot{y}} = m\dot{y} \equiv \text{const}$

Problem 4) $L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) - U(r)$; $\frac{\partial L}{\partial x_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) = 0$

There are no z and θ dependence, $\left. \begin{array}{l} \frac{\partial L}{\partial \dot{z}} = \text{const.} \\ \frac{\partial L}{\partial \dot{\theta}} = mr^2 \dot{\theta} \equiv \text{const.} \end{array} \right\}$

$\sin \theta$ is the tangential velocity,

$\Rightarrow mr \frac{\partial}{\partial (r\dot{\theta})} = m\dot{\theta}$ is the angular momentum

around the z -axis.

Problem 5) $L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) - U(r, \theta)$

where ϕ is the angle around the equator, and θ angle down from the north pole.

No ϕ dependence $\Rightarrow \frac{\partial L}{\partial \dot{\phi}} = mr^2 \sin^2 \theta \dot{\phi} \equiv \text{const.}$ (I)

$\left. \begin{array}{l} r \sin \theta \text{ is the distance from } z \text{ axis.} \\ r \sin \theta \dot{\phi} \text{ is the speed in the tangential direction around } z\text{-axis.} \end{array} \right\}$ (II)

(I), (II) $\Rightarrow mr^2 \sin^2 \theta \dot{\phi} \equiv \text{angular momentum around the } z\text{-axis.}$

Problem 6) $\frac{\partial L}{\partial t} = 0$; $\frac{dL}{dt} = \sum_j \frac{\partial L}{\partial \dot{q}_j} \dot{q}_j + \sum_j \frac{\partial L}{\partial \ddot{q}_j} \ddot{q}_j$ (I)

We also have $\frac{\partial L}{\partial \dot{q}_j} = \frac{d}{dt} \frac{\partial L}{\partial \ddot{q}_j}$ (II)

(I), (II) $\Rightarrow \frac{dL}{dt} - \sum_j \dot{q}_j \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} + \sum_j \frac{\partial L}{\partial \ddot{q}_j} \ddot{q}_j = 0$

$\Rightarrow \frac{d}{dt} \left(L - \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} \right) = 0 \Rightarrow L - \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} = -H \equiv \text{const.}$ (III)

Assuming the potential energy does not depend explicitly on $\dot{q}_j$,

$\frac{\partial L}{\partial \dot{q}_j} = \frac{\partial (T-U)}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}$ (IV)

(III), (IV) $\Rightarrow (T-U) - \sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = -H$

However, we also know that $\sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = 2T$ (easy to see in cartesian coordinates).

$\Rightarrow T + U = H \equiv \text{const.}$ where the function H is called

the Hamiltonian of the system, which is the total energy of the system if:

① kinetic energy $\propto \dot{q}_i^2$ in the generalized coordinate.

② potential energy does not depend on velocity.

Problem 7) a) The Lagrangian is the difference of potential and kinetic energies: $L = T - U = \frac{1}{2} M \dot{r}^2 + \frac{1}{2} m (\dot{r}^2 + (r\dot{\theta})^2) + Mgy(l-r)$

where we have assumed that the potential at table height is zero.

E-L equation requires: $\left. \begin{array}{l} \frac{\partial L}{\partial r} - \frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = 0 \Rightarrow (M+m)\ddot{r} = mr\dot{\theta}^2 - Mgy \\ \frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = 0 \Rightarrow \frac{d}{dt} (mr^2 \dot{\theta}) = 0 \end{array} \right\}$

b) Circular motion occurs when $\ddot{r} = \ddot{\theta} = 0$ or $r = \text{const.}$

$(M+m)\ddot{r} = mr\dot{\theta}^2 - Mgy = \frac{mv^2}{r} - Mgy \Rightarrow \boxed{\frac{mv^2}{r} = Mgy}$

$\Rightarrow$ For circular motion, the gravitational force of M must act as the required centripetal force.

Problem 8) x_1 : The horizontal coordinate of M .

x_2 : The horizontal coordinate of m .

$x_1 + x_2$: The relative horizontal distance between M, m .

$\Rightarrow h = (x_1 + x_2) \tan \theta$ which is the height fallen by m .

$\Rightarrow L = \frac{1}{2} M \dot{x}_1^2 + \frac{1}{2} m (\dot{x}_2^2 + \dot{h}^2) + mgh$ (The potential energy of M is constant)

The E-L eqns for x_1, x_2 yield: $\left. \begin{array}{l} M\ddot{x}_1 + m(\ddot{x}_1 + \ddot{x}_2) \tan^2 \theta = m g \tan \theta \\ m\ddot{x}_2 + m(\ddot{x}_1 + \ddot{x}_2) \tan^2 \theta = m g \tan \theta \end{array} \right\}$

$\Rightarrow \boxed{\ddot{x}_1 = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}}$

$M\ddot{x}_1 - m\ddot{x}_2 = 0$

$\Rightarrow \frac{d}{dt} (M\dot{x}_1 - m\dot{x}_2) = 0$

$\Rightarrow$ Linear momentum is constant

